# AP BIOLOGY TOPIC 6.2

***DNA Replication: Semiconservative Fork Mechanics & Enzymes***

**UNIT 6: GENE EXPRESSION & REGULATION (12–16% WEIGHT)**

IST-1.M | SEMICONSERVATIVE | 5′→3′ SYNTHESIS | OKAZAKI FRAGMENTS

Credits: mitali120501@gmail.com

## §1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 6.2 (DNA REPLICATION: SEMICONSERVATIVE FORK MECHANICS & ENZYMES)

*CED Core Vocabulary*

**Purpose** — Ensures faithful continuity of genetic information before cell division (mitosis/meiosis); occurs during S phase of interphase for growth, tissue repair, and...

**Meselson-Stahl Experiment (1958)** — Cultured E. coli in heavy nitrogen ( $^{15}N$ ) then shifted to light nitrogen ( $^{14}N$ ). After Gen 1 , 100% of DNA formed a single intermediate hybrid band ( $^{15}N/^{14}N$ ,...

**Why Primers Are Required** — DNA Polymerase cannot initiate a new polynucleotide chain from scratch — it requires Primase to lay down a short RNA primer that provides the initial free 3′-OH !

**Lagging Strand (5′ → 3′ Template)** — Synthesized discontinuously in short segments called Okazaki fragments moving away from the replication fork ; requires a new RNA primer for every fragment ,...

**Semiconservative Replication Model** — Parental double helix unwinds, and each original strand serves as a template for synthesizing a new complementary strand ( A↔T , G↔C ).

**Strict 5′ → 3′ Synthesis Rule** — DNA Polymerase can ONLY add incoming deoxynucleoside triphosphates (dNTPs) to a pre-existing free 3′-OH (hydroxyl) end of a growing strand.

**Leading Strand (3′ → 5′ Template)** — Synthesized continuously in one piece into (toward) the advancing replication fork ; requires only 1 RNA primer .

**Proofreading & Mismatch Repair** — DNA Polymerase checks each newly added base and uses 3′ → 5′ exonuclease activity to excise mismatched nucleotides immediately; mismatch repair enzymes and...

## §2. PURPOSE & SEMICONSERVATIVE NATURE OF DNA REPLICATION

*S Phase of Interphase*

**HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]**

**Purpose & Semiconservative Nature of DNA Replication** ➔ **Purpose** ➔ **Semiconservative Replication Model** ➔ **Meselson-Stahl Experiment (1958)**

- **Purpose:** Ensures faithful continuity of genetic information before cell division (mitosis/meiosis); occurs during **S phase of interphase** for growth, tissue repair, and reproduction.
- **Semiconservative Replication Model:**
  - Parental double helix unwinds, and **each original strand serves as a template** for synthesizing a new complementary strand ( **A↔T** , **G↔C** ).
  - Each resulting daughter DNA molecule consists of **1 old (parental) strand + 1 newly synthesized strand**.
- **Meselson-Stahl Experiment (1958):** Cultured *E. coli* in heavy nitrogen ( **$^{15}N$** ) then shifted to light nitrogen ( **$^{14}N$** ). After **Gen 1**, 100% of DNA formed a single intermediate hybrid band ( **$^{15}N/^{14}N$** , ruling out conservative); after **Gen 2**, 50% hybrid ( **$^{15}N/^{14}N$** ) + 50% light ( **$^{14}N/^{14}N$** , ruling out dispersive).

**[CORE AXIOM / REMEMBER]** **Semiconservative = "Half Old, Half New":**
Every replicated double helix preserves one intact parental template strand paired with one newly polymerized daughter strand.

## §3. 5′ → 3′ DIRECTIONALITY & LEADING VS. LAGGING STRANDS

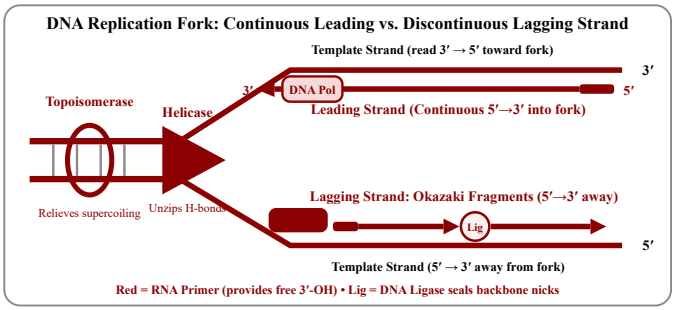
*Core AP Concept*

- **Strict 5′ → 3′ Synthesis Rule:**
  - **DNA Polymerase** can ONLY add incoming deoxynucleoside triphosphates (dNTPs) to a pre-existing free **3′-OH (hydroxyl) end** of a growing strand.
  - Therefore, **new DNA is ALWAYS synthesized in the 5′ → 3′ direction** while DNA Polymerase reads the **template strand in the 3′ → 5′ direction**.
  - **Why Primers Are Required:** DNA Polymerase *cannot initiate* a new polynucleotide chain from scratch — it requires **Primase** to lay down a short **RNA primer** that provides the initial free **3′-OH**!
- **Because DNA is Antiparallel, the Two Strands Replicate Differently at a Replication Fork:**
  - **Leading Strand (3′ → 5′ Template):** Synthesized **continuously** in one piece **into (toward) the advancing replication fork**; requires only **1 RNA primer**.
  - **Lagging Strand (5′ → 3′ Template):** Synthesized **discontinuously** in short segments called **Okazaki fragments** moving **away from the replication fork**; requires a **new RNA primer for every fragment**, which are later replaced with DNA and covalently sealed by **DNA Ligase**.

**[COMMON AP EXAM FRQ TRAP]** **Both Strands Are Synthesized 5′ → 3′:**
Never say the lagging strand is synthesized 3′ → 5′! Both leading and lagging strands are built strictly **5′ → 3′**; the lagging strand simply builds short 5′ → 3′ Okazaki fragments oriented away from the opening fork.

## §5. THE DNA REPLICATION FORK VISUALIZED

*All Enzymes Labeled*




## §6. MASTER TABLE OF DNA REPLICATION ENZYMES

*Must-Know CED Roles*

| ENZYME / PROTEIN | SPECIFIC MOLECULAR FUNCTION AT THE REPLICATION FORK |
|---|---|
| **Topoisomerase (DNA Gyrase)** | Binds ahead of the replication fork; transiently nicks and reseals the sugar-phosphate backbone to **relieve torsional strain and supercoiling** caused by unwinding. |
| **DNA Helicase** | **Unwinds the parental double helix** at the origin of replication by breaking **hydrogen bonds** between complementary base pairs. |
| **Single-Strand Binding Proteins (SSBPs)** | Coat separated single DNA strands to prevent them from snapping back together (re-annealing) or being degraded. *(Helpful context; not tested by name alone)* |
| **Primase** | An RNA polymerase that synthesizes a short complementary **RNA primer (5–10 nt)** to provide a **free 3′-OH group** for DNA Polymerase to start. |
| **DNA Polymerase (III & I)** | **Elongates new DNA strands 5′ → 3′ ONLY** by adding nucleotides to the 3′-OH end; proofreads new bases; replaces RNA primers with DNA nucleotides. |
| **DNA Ligase** | Catalyzes covalent **phosphodiester bonds** to seal nicks in the sugar-phosphate backbone, joining **Okazaki fragments** into a continuous strand. |

**[CORE AXIOM / REMEMBER]** **5-Word Enzyme Summary: Topoisomerase** = untwists strain • **Helicase** = unzips H-bonds • **Primase** = lays first brick (RNA primer) • **DNA Polymerase** = builds 5′→3′ • **Ligase** = glues fragments!

**[AP EXAM FRQ SCORING STRATEGY]** **Predict the Mutation / Inhibitor**

**Question:** AP FRQs love asking "What happens if Enzyme X is mutated or inhibited?"

- **Mutated Helicase:** Strands cannot separate → no replication bubble forms.
- **Mutated Primase:** No 3′-OH primer → neither leading nor lagging strand can start.
- **Mutated DNA Ligase:** Leading strand finishes normally, but the **lagging strand remains in unjoined Okazaki fragments**!

## §4. REPLICATION ACCURACY, PROOFREADING & TELOMERES

*High Fidelity*

**HIERARCHICAL CONCEPT & PATHWAY MAP** **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

**Replication Accuracy, Proofreading & Telomeres** ➔
**Proofreading & Mismatch Repair** ➔ **End-Replication Problem in Eukaryotes**

- **Proofreading & Mismatch Repair:** **DNA Polymerase** checks each newly added base and uses *3′ → 5′ exonuclease activity* to excise mismatched nucleotides immediately; mismatch repair enzymes and *nucleotide excision repair* (e.g., fixing UV thymine dimers) keep error rates near **~1 in $10^{10}$** bp.
- **End-Replication Problem in Eukaryotes:** Linear chromosomes shorten at 5′ lagging ends when terminal RNA primers are removed; non-coding repetitive **telomeres** (**TTAGGG**) buffer coding genes, and **telomerase** extends telomeres in germ/stem cells (and cancer cells).

## §7. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

*High-Yield Exam Rigor*

**MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 6.2** **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

**Topic 6.2: DNA Replication: Semiconservative Fork Mechanics & Enzymes** ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔
**Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

| SCHOLARLY DIMENSION | COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION |
|---|---|
| **I. Experimental Design & Required Lab Controls** | AP Investigations 8 & 9 (pGLO Bacterial Transformation & Restriction Enzyme Analysis of DNA): Heat-shock transformation of E. coli with pGLO plasmid (bla ampicillin-resistance gene + GFP under arabinose-inducible araC promoter). Plate comparisons: −pGLO/LB (lawn, viability control), −pGLO/LB+Amp (no growth, antibiotic lethality control), +pGLO/LB+Amp (white colonies, transformation), +pGLO/LB+Amp+Ara (glowing green colonies under UV, inducible gene expression). **Null Hypothesis ($H_0$):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone. |
| **II. Quantitative Formulas & Graph Interpretation** | Quantitative & Graph Analysis: Transformation Efficiency = (Total colonies counted on selective plate) / (µg of plasmid DNA spread). Gel Electrophoresis Semi-Log Standard Curve: DNA fragment size (bp, $\log_{10}$ scale on y-axis) vs. migration distance (mm, linear scale on x-axis) yields a straight line for interpolating unknown fragment lengths. PCR amplification after n thermal cycles yields $2^n$ target DNA copies. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject $H_0$). |
| **III. Pathology, Mutation & Disruption Case Study** | Clinical & Molecular Disruption Case Study: Splice-site mutations in the β-globin gene (β-Thalassemia) prevent the spliceosome (snRNPs) from excising an intron or cause exon skipping, shifting the mRNA reading frame and introducing a premature nonsense stop codon (UAA/UAG/UGA). In oncology, epigenetic hypermethylation of CpG islands in tumor-suppressor promoters (e.g., BRCA1 or MLH1) or histone deacetylation condenses euchromatin into transcriptionally silent heterochromatin without altering the primary DNA sequence. |
| **IV. AP FRQ Task Verb Scoring Templates** | • **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure.<br>• **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."*<br>• **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed.<br>• **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 6.2 to prove why the claim holds. |



Support this site
Scan to donate